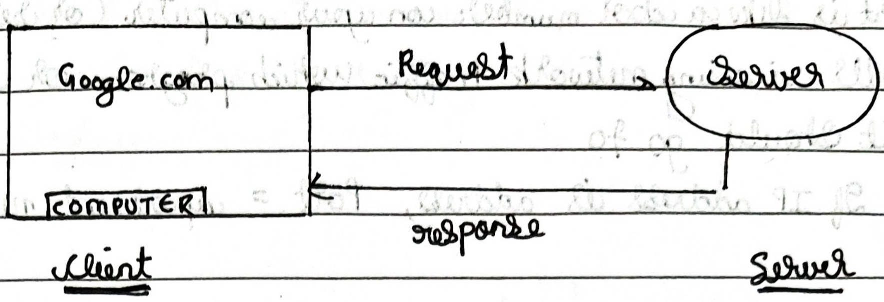
Phase - 1

Networking



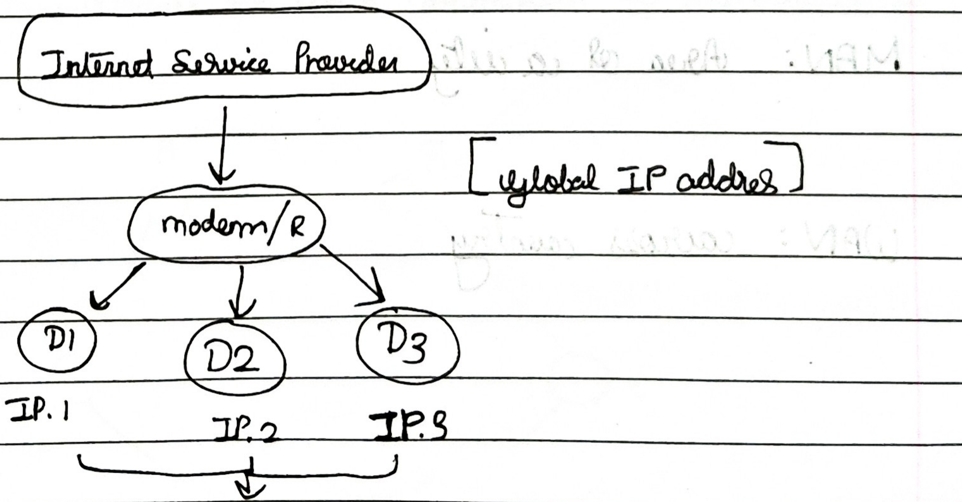
Protocols

- TCP - Transmission control proctocol - email, vid, photos
  - Your data arrives at correct order, non-missing at final destination

- UDP - User datagram protocol - Ex: video calls, streaming
  - Speed over reliability, doesn't care if packet data is lost.

- HTTP: Hypertext ~~Ma~~Transfer protocol - Ex: APIs, websites
  - language of web browsers & servers on to communicate each other

Data is transffered?



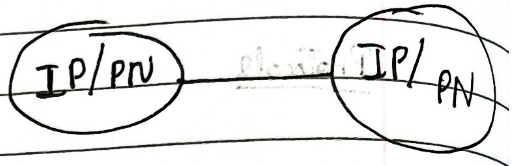
Same IP address [local IP address]

done by DHCP [Dynamic Host Control Protocol]

Port

Port is like a door number on your computer (or server) that tells incoming network traffic which program or service it should go to

If IP address is address, Port = apartment number



- Ports are 16 bit no.

$\Rightarrow 2^{16}$ // [65536]

0-1023 (reserve ports)
1024-49152 [application]

How connected?

Physical : optical fibres, cables
Wireless : bluetooth, wi-fi

LAN : Small house / building / corporation
[Ethernets

MAN : Area of a city

WAN : across country

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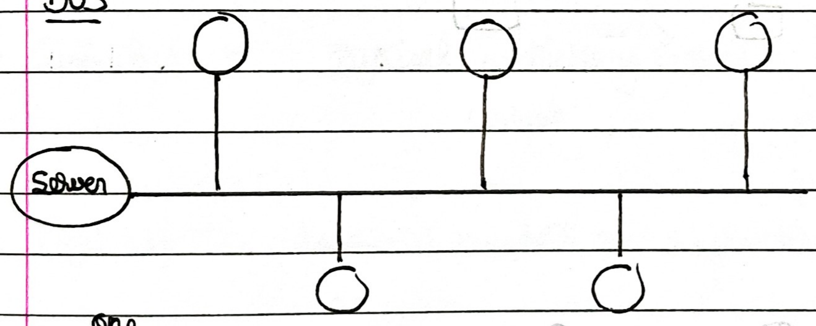
* SONET [Synchronous Optical network]
high speed, fiber-optic communication standard used to transmit huge amounts of data over long distance
from relay

* MODEM [Modulator demodulator]
• Connects you to ISP
• Converts digital data to analog signals for transmission and vice versa

* Router : distributes internet connection from modem to multiple devices

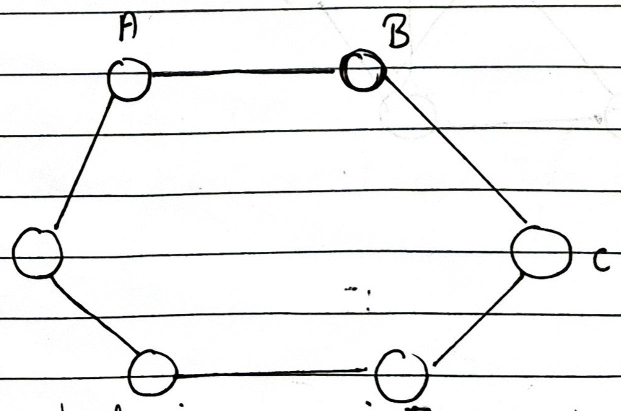
Topologies

① BUS



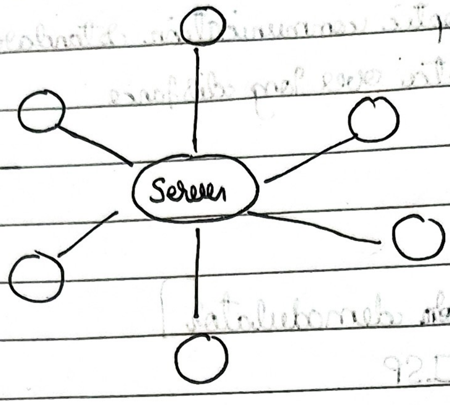
one host/main channel for data transmission & communication

② Ring



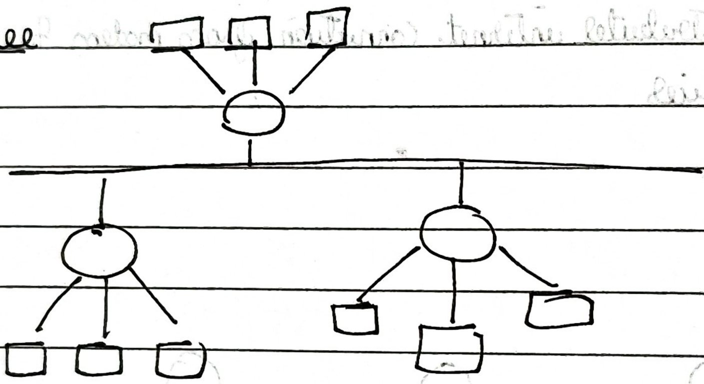
here each device communication happens

(3) Star

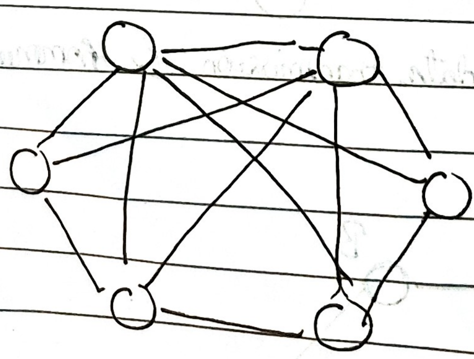


Connected to one main server

4) Tree



5) Mesh



Expensive & robust

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- **Structure of Network**

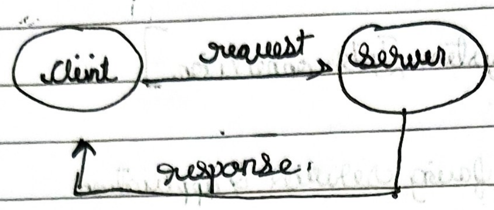
OSI Model [open system Interconnection]
↓

		Ex Protocols
• Application →	User-facing services & application	HTTP, FTP, SMTP
• computer		
• Presentation →	Data formatting, encryption, compression	SSL/, JPEG, MP3
• Session →	Starts, maintains & ends communication session	NetBios, RPC
• Transport →	reliable delivery, error check segmentation	TCP, UDP
• Network →	Logical addressing & routing	IP, ICMP, Routers
• Data Link →	Physical addressing (MAC), error	
• Physical →	Transmits raw bits over physical medium	Cables, wi-fi

TCP/IP model
developed by ARPANET

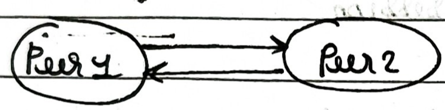
- Application layer
- user interacts here, consist application [Ex: whatsapp,
- lies on devices
- Protocols

- Client Server-Architecture



- Peer to peer

there, every device (peer) on network act as both client & a server.



- Protocols

web protocols

TCP/IP:

* HTTP
* DHCP
* FTP
* SMTP
* POP3
* SSH
* VNC

* Telnet:

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Ports

Ephemeral ports : assign itself random port number.

→ **HTTP**

$$\text{Client} \underset{\text{Response}}{\overset{\text{Req}}{\rightleftarrows}} \text{Server}$$

- uses TCP inside.
- HTTP is a stateless

- Method

① ~~Post~~ GET
② POST
③ PUT
④ DELETE

Error Codes / Status

1XX - Informational codes.
2xx - success
3xx - Redirection
4xx - client Error
5xx - Server Error

- **Cookies**

Unique string, which is stored in a browser.

- Third party cookies

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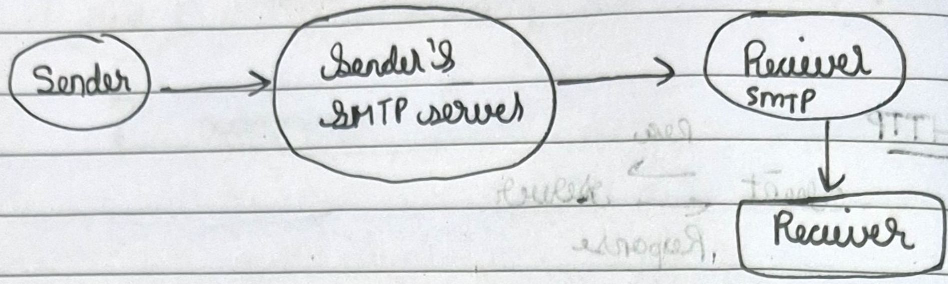
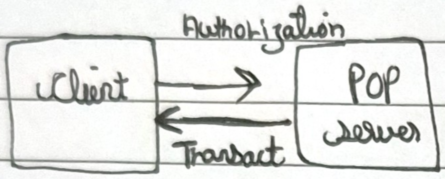
How E-mail works

Uses SMTP

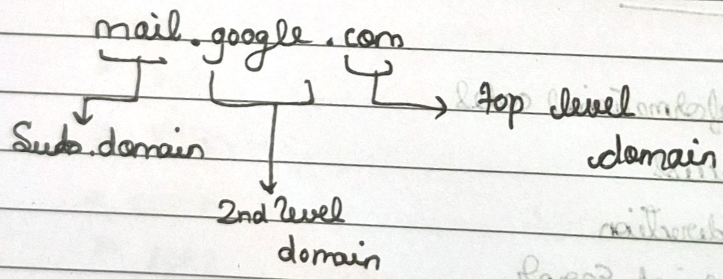


POP- Post office protocol [Port 110]



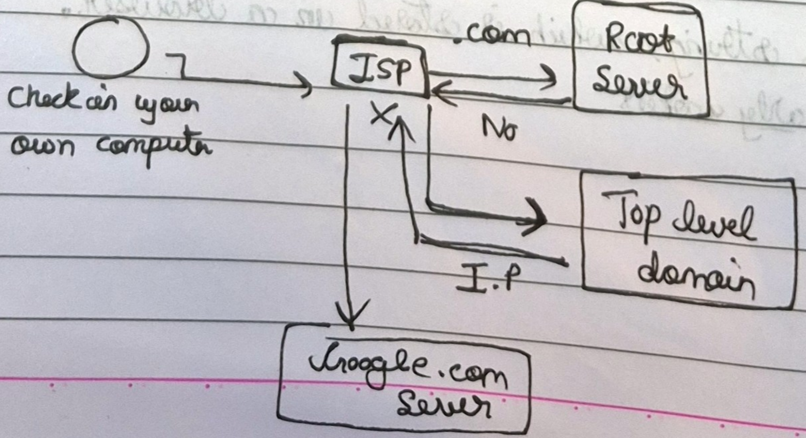
- Download & keep email on servers or delete it.

- Domain Name System (DNS) [like a phonebook of Internet]



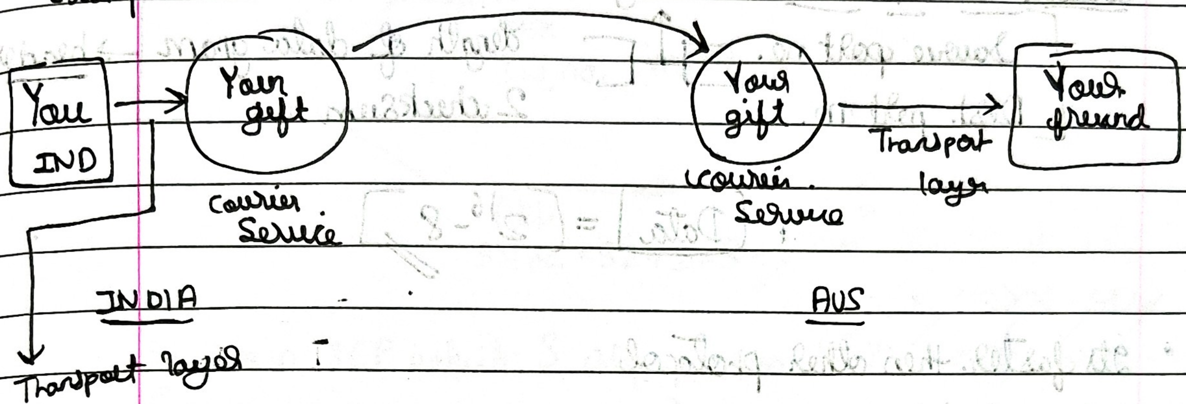
Example Chart

Type Google.com



2) Transport layer [end to end communication between applications on different devices]

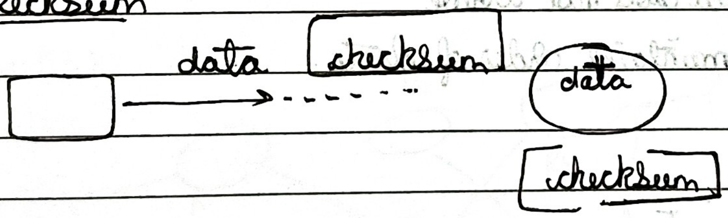
Example



* uses TCP/IP

- TCP: takes care of congestion control
  - congestion control algorythm built in TCP

Checksum



Checksum is an error detection technique where a numeric value is calculated from data and compared at sender and receiver to ensure data integrity

Timers

Timers is a countdown mechanism in networking protocol used to manage timeouts, retransmissions, and connection reliability.

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UDP - User datagram protocol
- Data may or may not be delivered
- Data may change
- Data may not be in order

inside UDP Packet

8 bytes

[Source port no. ; Dest. port no.] [length of data gram ; 2 checksum] → header

$\boxed{\text{Data}} = [2^{16} - 8]$

- Its faster then other protocol
- Video Conferenceing
- DNS uses UDP

TCP [Transmission control protocol]
- Congestion control
  - ↳ data does not arrive
  - ↳ maintains order of data

Feature
- connection oriented
- Error control
- Congestion control
- Full Duplex

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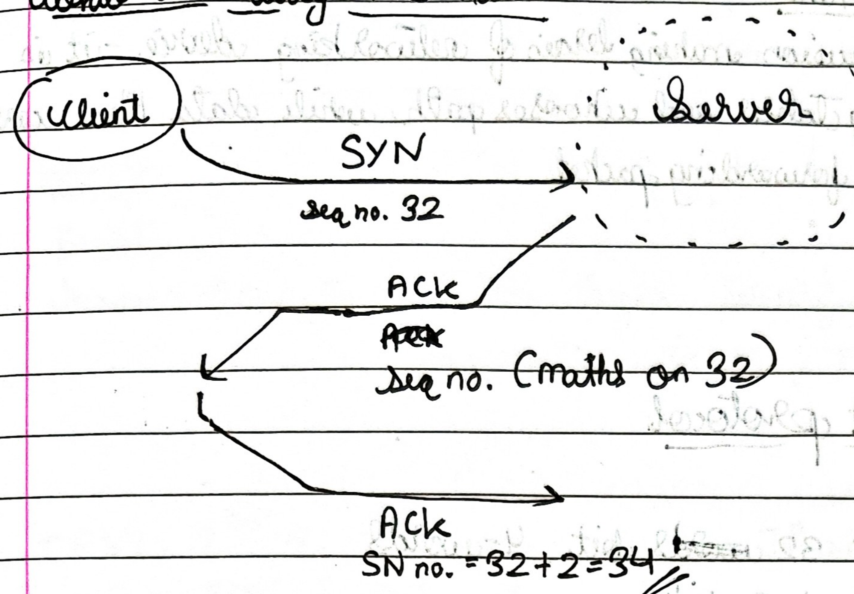
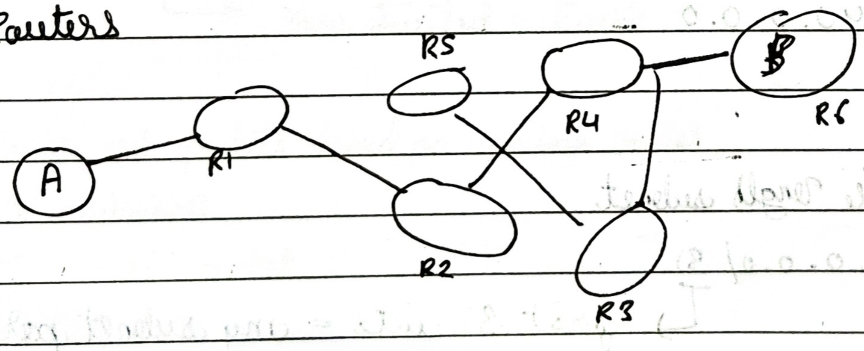
What is 3-way handshake



- It is in TCP which 3 step process (SYN → ~~ACK / ACK~~ SYN-ACK → ~~SYN~~ ACK)
(SYN → ACK → ACK → ACK)
to establish reliable internet connection

- **Network layer** [RIP, OSPF, BGP]
→ here we work with router

Routers



→ Chooses best path to send packets in destination using routing & forwarding table

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Control Plane

is a decision making brain of networking device - it is built on routing Tables and chooses path, while data Plane carries out the forwarding packet

Internet protocol

IPV4 → 32 ~~words~~ bit, 4-words

IPV6 → 128 bit

Class of IP address

A - 0.0.0.0

B - 128.0.0.0

C - 192.0.0.0

D - 224.0.0.0

E - 240.0.0.0

variable length subnet

12.0.0.0/31

↳ first 31 bits = my subnet part

$\underbrace{192.0.1}_{\text{it will be same}}.0/24$

Internet Engineering task force [assign handles IP address in devices & ISP]

~~Packets~~

Packets

header - 20 bytes → IPV6, length, identification no., protocols, address, TTL

IPV4 : $2^{32} \approx$ 4.3 billion unique IP address

IPV6 : $2^{32 \times 4} = 2^{128} = 3.4 \times 10^{38}$

Cons

* Not Backward Compatible
* ISP would have to shift, lot of hardware work.

Middle box

1) Firewall → global network
→ Your trusted network

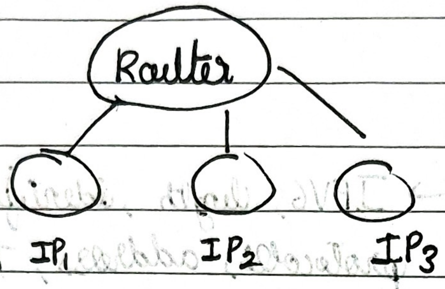
Filter out packets based on various rules

→ addresses
→ modify packet
→ Port nos
→ Flags
→ protocols

2) Network address translation - Maps private IPs to public IPs

Datalink layer

DHCP



If new device allocated → DHCP server [Pool of IP addresses]

~~AB~~ ARP- when devices doesn't know the ~~route~~ data link layer of dest. device the it will ask in all available devices.

[Address resolution protocol]

Frame : Data link layer address of sender & IP address of dest.